

To find the following the machine learning regression method using in r2 value

1. Multiple LINEAR REGRESSION(R2 Value) = 0.9358

2. SUPPORT VECTOR MACHINE:

S.no	Hyper Parameter	Linear (r2_score)	RBF (Non Linear r2_score)	Poly (r2_score)	Sigmoid(r2 Score)
1	C0.01	-0.0570	-0.0575	-0.0500	-0.0500
2	C10	-0.0396	-0.0568	-0.0500	-0.0500
3	C100	0.1065	-0.0507	-0.0100	-0.0300
4	C1000	0.7803	0.0068	0.2600	0.1800
5	C2000	0.8767	0.0675	0.4800	0.3900
6	C5000	0.9004	0.2100	0.7900	0.7300

The SVM Regression use R2 value(Linear and hyper parameter (C5000) = 0.9004)

3. DECISION TREE:

S.NO	criterion	MAX_FEATURE	SPLITTER	R VALUE
1	MSE	Auto	Best	0.9087
2	MSE	Auto	Random	0.6642
3	MSE	Sqrt	Random	0.7055
4	MSE	Sqrt	Best	0.4218
5	MSE	Log2	Random	0.7055
6	MSE	Log2	Best	0.4218
7	friedman_mse"	Auto	Best	0.9087
8	friedman_mse"	Auto	Random	0.6642
9	friedman_mse"	Sqrt	Random	0.7055
10	friedman_mse"	Sqrt	Best	0.4218
11	friedman_mse"	Log2	Random	0.7055
12	friedman_mse"	Log2	Best	0.4218
13	absolute_error	Auto	Best	0.9329
14	absolute_error	Auto	Random	0.7275
15	absolute_error	Sqrt	Random	-1.3619
16	absolute_error	Sqrt	Best	0.6033
17	absolute_error	Log2	Random	-1.3619
18	absolute_error	Log2	Best	0.6033
19	poisson	Auto	Best	0.9134
20	poisson	Auto	Random	0.6642
21	poisson	Sqrt	Random	0.7055
22	poisson	Sqrt	Best	0.2589
23	poisson	Log2	Random	0.7055
24	poisson	Log2	Best	0.2589

The DECISION TREE Regression use R2 value(Best hyper parameter is absolute_Error,Auto,Best = 0.9329)

